UNIVERSITY OF MINES AND TECHNOLOGY, TARKWA

FIRST SEMESTER EXAMINATIONS, NOV. – DEC. 2019

COURSE NO: **CE/EL 465**
COURSE NAME: **ARTIFICIAL INTELLIGENCE**
CLASS: **CE/EL IV** **TIME: 3 HRS**

Name**:** _____Index Number: _____

Circle the correct answer from the choices provided for each of the following questions

1. Which function will select the lowest expansion node at first for evaluation?
   a. Greedy best-first search
   b. Depth-first search
   c. Breadth-first search
   d. Both A and B

   Correct answer: Best first search
2. Which search uses problem-specific knowledge beyond the definition of the problem?
   a. Informed search
   b. Depth-first search
   c. Best-first search
   d. Uninformed
3. A* algorithm is based on
   a. Breadth-first search
   b. Depth-first search
   c. Bulkworld problem
   d. Best-first search
4. Which is not a property of knowledge representation?
   a. Representational verification
   b. Representational adequacy
   c. Acquisitional efficiency
   d. Inferential efficiency
5. A bi-directional feedback loop links computer modelling with
   a. Artificial science
   b. Heuristic processing
   c. Human intelligence
   d. Cognitive science
6. A process that is repeated, evaluated and re-defined is called
   a. diagnostic
   b. descriptive
   c. interpretative
   d. iterative
7. Which of the following is not done by a Goal-based agent?
   a. Conclusion
   b. Planning
   c. Searching
   d. Inference
8. Neural Networks are complex …… with many parameters.

a. Linear Functions
b. Exponential Functions
c. Nonlinear Functions
d. Discrete Functions

9. A perceptron is a ……..
a. Feed-forward neural network
b. Feed Forward-backward algorithm
c. Back-propagation algorithm
d. Back-tracking algorithm

10. An algorithm is complete if
a. It starts with a solution
b. It does not terminate with a solution
c. It terminates with a solution when one exists
d. It has a loop

11. The action 'STACK(A, B)' of a robot arm specify to
a. Place block B on Block A
b. Place blocks A, B on the table in that order
c. Place blocks B, A on the table in that order
d. Place block A on block B

12. Which of the following, is a component of an expert system?
a. inference engine
b. user interface
c. knowledge base
d. all of the mentioned

13. The following task/tasks Artificial Intelligence could not do yet
a. Understand natural language robustly
b. Web mining
c. Construction of plans in real time dynamic systems
d. All of the mentioned

14. Which particular generation of computers is associated with artificial intelligence?
a. Second
b. Fourth
c. Fifth
d. Third

15. In which of the following situations might a blind search be acceptable?
a. real-life situation
b. complex game
c. small search space
d. all of the mentioned

16. Computers normally solve problem by breaking them down into a series of yes-or-no decisions represented by 1s and 0s. What is the name of the logic that allows computers to assign numerical values that fall somewhere between 0 and 1?
a. Human logic
b. Boolean logic
c. Fuzzy logic
d. Operational logic

17. In language understanding, the levels of knowledge that does not include
a. Empirical
b. Phonological

c. Syntactic
d. Semantic

18. In Baye's theorem, what is the meant by P(Hi|E)?

a. The probability that hypotheses Hi is true given evidence E
b. The probability that hypotheses Hi is false given evidence E
c. The probability that hypotheses Hi is true given false evidence E
d. The probability that hypotheses Hi is false given false evidence E

19. Which search strategy among the following makes the most efficient use for memory

a. Depth-first search
b. Breadth-first search
c. Linear search
d. Optimal search

20. What is state space?

a. The whole problem
b. Your definition to the problem
c. A space where you know the solution
d. Representation your problem with variable and parameter

21. Face recognition system is based on

a. applied AI
b. parallel AI
c. serial AI
d. strong AI

22. The performance of an agent can be improved by

a. Learning
b. Observing
c. Perceiving
d. None of the mentioned

23. External actions of the agent are selected by

a. Perceive
b. Performance
c. Learning
d. Actuator

24. A completely automated chess engine (Learn from previous games) is based on

a. Strong Artificial Intelligence approach
b. Weak Artificial Intelligence approach
c. Cognitive Artificial Intelligence approach
d. Applied Artificial Intelligence approach

25. What kind of environment is strategic in artificial intelligence?

a. Deterministic
b. Rational
c. Partial
d. Stochastic

26. Agents behavior can be best described by

a. Perception sequence
b. Agent function
c. Sensors and Actuators

d. Environment in which agent is performing

27. Which search is implemented with an empty first-in-first-out queue?

a. Depth-first search
b. Breadth-first search
c. Bidirectional search
d. None of the mentioned

28. Strategies that know whether one non-goal state is "more promising" than another are called

a. Informed & Unformed Search
b. Unformed Search
c. Heuristic & Unformed Search
d. Informed & Heuristic Search

29. LIFO is ……. whereas FIFO is ……..

a. Stack, Queue
b. Queue, Stack
c. Priority Queue, Stack
d. Stack, Priority Queue

30. What is the heuristic function of greedy best-first search?

a. f(n) != h(n)
b. f(n) < h(n)
c. f(n) = h(n)
d. f(n) > h(n)

31. Which search is complete and optimal when h(n) is consistent?

a. Depth-first search
b. Best-first search
c. Both Best-first & Depth-first search
d. A* search

32. Which search method will expand the node that is closest to the goal?

a. Best-first search
b. A* search
c. Greedy best-first search
d. None of the mentioned

33. In A* approach evaluation function is

a. Heuristic function
b. Path cost from start node to current node
c. Path cost from start node to current node + Heuristic cost
d. Average of Path cost from start node to current node and Heuristic cost

34. In generic graph search, a set of nodes generated and yet to be explored during a search is known as the

a. Child nodes
b. Parent nodes
c. Frontier
d. None of the mentioned

35. In AI problem definition, the property that describes the result of each actions taken in a particular state is known as

a. Transition model
b. Percept sequence

c. Goal Test

d. -

e. Initial state

What is perceptron? A single layer feed-forward neural network with pre-processing

A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear with the constant of proportionality being equal to 2. The inputs are 4, 10, 5 and 20 respectively. What will be the output? The output is found by multiplying the weights with their respective inputs, summing the results and multiplying with the transfer function. Therefore: Output = 2 * (1*4 + 2*10 + 3*5 + 4*20) = 238.

What is back propagation? It is the transmission of error back through the network to allow weights to be adjusted so that the network can learn

Who among the following is regarded as the "father" of modern artificial intelligence? John McCarthy

Fill the blank spaces provided in the questions below with the most appropriate answers

36. ……Turing Test ……. is a proposed means of testing the intelligence of a machine by ……Alan M. Turing …….

36. Search algorithm ……. find solution for a program by trying different sequences of actions/operators until a solution is found.

38. In AI, the representation of a problem in terms of states and operators that change states in known as …State Space.

39. ..Depth-first search…implements stack operation for searching …the states……

40. Two other names of the uninformed search strategy are ……Blind search …… and ……unguided search……

41. The area of AI that investigates methods of facilitating communications between humans and computers is called ……Natural Language Processing…….

42. ……Cognitive Science… is a field that investigates the mechanics of human intelligence.

43. The Professor at the Stanford University that coined the word 'artificial intelligence' in 1956 at a conference held at Dartmouth college is ……John McCarthy……

44. The embodiment of human intellectual capabilities within a computer is the category AI called …Strong Artificial Intelligent……

45. The FIFO queue strategy is often employed to implement …… Breadth-first search

46. Natural language processing is divided into two sub-fields namely …Understanding… and …Generation…

47. The conference that launched the AI revolution in 1956 was held at …Dartmouth....is the Machine learning autonomous acquisition of knowledge through the use of computer programs.

48. The conference that launched the AI revolution in 1956 was held at …Dartmouth….

49. …Heuristic ……is a term is used for describing the judgmental or common-sense part of problem solving.

50. The action of the Simple reflex agent completely depends upon …Current Perception…...

51. In terms of observability, the crossword puzzle can be classified as …Fully Observable...

52. An agent consists of architecture and …Program...

53. A search algorithm takes …Problem... as an input and returns …Solution... as an output.

54. The …Travelling Salesman Problem…….. is a touring problem in which each city must be visited exactly once. The aim is to find the shortest tour.

55. The best strategy to solve a game playing problem is by …Heuristic Approach...

56. Uniform-cost search expands the node n with the…Lowest Path Cost...

57. The informed search strategy is also known as …Heuristic Search...

58. The …Binary Tree... is a special tree structure in which each node branches into exactly two paths below it.

59. In the tree structure, the ..Height.. of a node is the length of the path from the root to the node.

<u>Using T or F respectively, indicate whether the following statements are True or False. Note: Wrong guesses shall be penalized, take the 5th if you are not sure</u>

60. Artificial intelligence is the study of how to make computers do things which, at the moment, people can do better. True

61. Artificial intelligence does not necessarily imply understanding. False

62. A heuristic is a way of trying to discover something or an idea embedded in a problem. True

63. Natural languages are context-oriented free. False

64. In an Unsupervised learning specific output values are not given. True

65. If a hypothesis says it should be positive, but in fact it is negative, we call it true negative hypothesis. False

66. A statement is satisfiable if there is some interpretation for which it is true. True

67. The intersection two context-free languages is context-free. True

68. The entire tree so far been generated must be stored in BFS. True

69. Good data structures are likely to affect the performance of learner system. False
70. Perception involves sights, sounds, smell, touch and acting. False
71. A natural language generation program must decide what to say. True
72. The action of the Simple reflex agent completely depends upon perception history. False
73. A basic line following robot is based on is based on weak artificial intelligence approach. True
74. Humans are examples of the intelligent agent/agents. True
75. Performance Measures are fixed for all agents. True
76. An omniscient agent knows the actual outcome of its actions and can act accordingly; but omniscience is impossible in reality. Rational Agent always does the right thing; but Rationality is possible in reality. True
77. The game of Poker is a single agent. False
78. A problem in a search space is defined by initial state. True
79. Solution quality is measured by the path cost function, and an optimal solution has the highest path cost among all solutions. False
80. There is always a recursive breadth-first search equivalent to the depth-first search strategy. False
81. Depth-first search guarantees optimal solution because no viable solution is omitted. False
82. Both stacks and queues can be implemented as linked list. True
83. Forward chaining is query driven while backward chaining is data driven. false
84. The run time of an algorithm is the time taken for the algorithm to solve a decision problem. True

THEORY

You are trained to be an Engineer not a writer, answer Question 1 any other ONE of the remaining questions precisely

1. **The Monkey and the Banana Problem**

   A hungry monkey finds itselt in a room in which a bunch of bananas is hanging from the ceiling. The monkey unfortunately cannot reach the bananas. However, in the room there are also a chair and a stick. The ceiling is just the right height so that a monkey standing on a chair could knock the bananas down with the stick. The monkey knows how to

move around, carry things around and wave a stick in the air. The monkey needs the best sequence of steps to take in order to acquire its lunch.

a) Describe the PEAS

b) Formulate the problem in terms of state space.

2. a) Distinguish between strong artificial intelligence and weak artificial intelligence giving one example in each case

   b) Define an expert system and give one example.

   c) What is uniform cost search?

3. a) How is an uninformed search method different from an informed search method?

   b) Mention ten application areas in AI.

   c) State and explain the three Artificial Intelligence problem domains.

Course Instructor

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